TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Determination of the Focal Length of a Convex Lens by the Conjugate Foci Method

Object and image distances measured on an optical bench, pair by pair — every pair must satisfy 1/f = 1/v + 1/u, and swapping u and v gives the same image: conjugacy itself.

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1. Aim and theory

Aim: to find the focal length f of a convex lens from pairs of conjugate distances. For a thin lens forming a real image (Physics-I Topic 6):

$$1/f = 1/v + 1/u \;\Rightarrow\; f = uv/(u + v)$$

with u the object distance and v the image distance from the lens. Object and image are **conjugate**: a pair (u, v) that focuses can be interchanged — put the object where the image was and the image forms where the object was (reversibility of light). The bench turns this into data: many (u, v) pairs, each yielding f, plus two graphical routes — the u–v graph (whose points lie on a hyperbola with asymptotes at u = v = f… precisely, the line u = v meets the curve at u = v = 2f) and the 1/u–1/v plot, whose intercepts are both 1/f.

2. Apparatus and procedure

- **Apparatus:** optical bench with uprights, convex lens in a holder, an illuminated object (cross-slit or arrow), a screen, metre scale along the bench.
- Find f roughly (image of a distant object on the screen) to plan the range. Correct for bench error first if the uprights' indices are offset (index correction).
- Set the object at a chosen u > f; move the screen until a sharp inverted image forms; read the lens and screen positions; v is their difference. Record the image size too (magnification m = v/u is a useful cross-check).
- Repeat for a series of u values, from just above f to the far end of the bench; include points on both sides of u = 2f.

- Compute f from each pair and average; verify with the 1/u–1/v graph (straight line, intercepts 1/f on both axes).

3. Worked calculation (model data)

RESULT (EXTRACT OF THE TABLE)

U (CM)	V (CM)	F = UV/(U+V) (CM)	M = V/U
25.0	100.0	20.0	4.00 (magnified)
30.0	60.0	20.0	2.00
40.0	40.0	20.0	1.00 (same size)
60.0	30.0	20.0	0.50
100.0	25.0	20.0	0.25 (diminished)

Mean f = **20.0 cm**. Read the table's lessons: at u = v = 2f the image is same-size; conjugacy appears openly — (25, 100) and (100, 25) are the same experiment read backwards, as are (30, 60) and (60, 30).

4. Precautions

- The lens, object and screen centres must be at the same height and the lens plane vertical and perpendicular to the bench, or distances are not measured along the axis.
- Judge sharpness by bracketing: approach the best focus from both sides and take the midpoint of the "acceptably sharp" range.
- Apply index (bench) correction: measure the true lens-to-screen distance with a rod if the upright indices do not mark the true planes.
- Keep u > f always (at u ≤ f no real image exists to catch on a screen).
- Avoid very large u/v ratios: the image becomes too dim or too small to focus sharply.

5. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. What does "conjugate foci" mean, and which principle of optics guarantees the pairing?
2. An object at 45.0 cm from a lens images at 22.5 cm. Find f, and state where the image of an object placed at 22.5 cm would form.

3. Why does the u = v point give f immediately, and what is the magnification there?
4. In the 1/u–1/v graph, what do the two intercepts give, and what should the line's slope be?
5. Why is no real image obtained for u < f, and how is that configuration used in a magnifying glass?
6. A student's five (u, v) pairs give f values 19.2, 19.8, 20.1, 20.0, 23.5 cm. What should they do before averaging?

Answer key

1. Conjugate foci are pairs of points so related that an object at either forms its image at the other. The guarantee is the reversibility of light: any ray path through the lens can be traversed in the opposite direction, so object and image positions are interchangeable.
2. f = (45.0 × 22.5)/(45.0 + 22.5) = 1012.5/67.5 = 15.0 cm. By conjugacy, an object at 22.5 cm images at 45.0 cm.
3. At u = v, the lens formula gives 1/f = 2/u, i.e. u = v = 2f, so f = u/2 directly from one setting. The magnification m = v/u = 1: a same-size, inverted image — the setting used in photocopiers and process cameras.
4. Each intercept equals 1/f (setting the other variable to infinity). The line has slope −1 (it is 1/v = 1/f − 1/u); a line through the data that is not at −1 exposes a systematic error such as an uncorrected bench index error.
5. For u < f the refracted rays diverge on the far side and never meet: the image is virtual, on the object's side, erect and magnified — precisely the magnifying glass, where the eye intercepts the diverging rays and sees the virtual image.
6. Interrogate the 23.5 cm outlier before averaging: check its arithmetic, then its raw readings (a mis-read scale or an unfocused image). Averages are for concordant data; a single wild point usually marks a blunder to find and repeat, not a number to dilute into the mean.

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